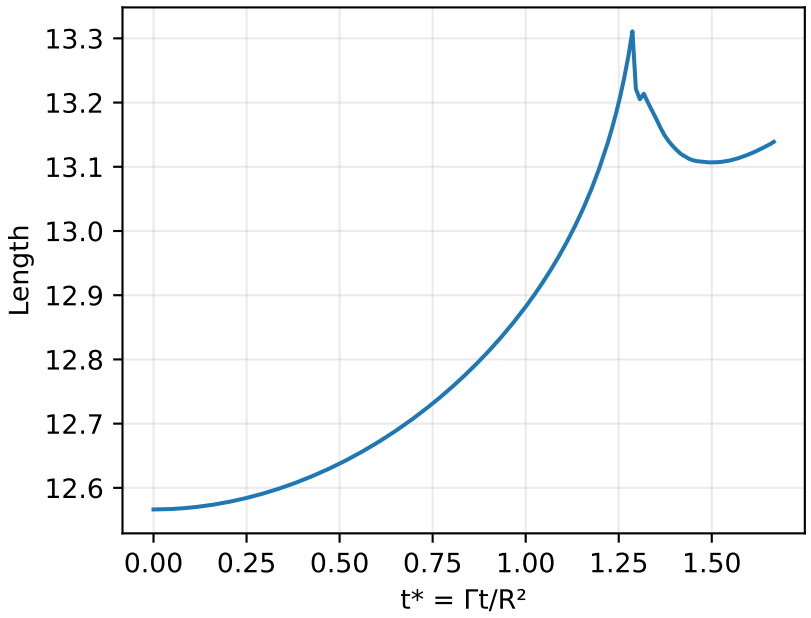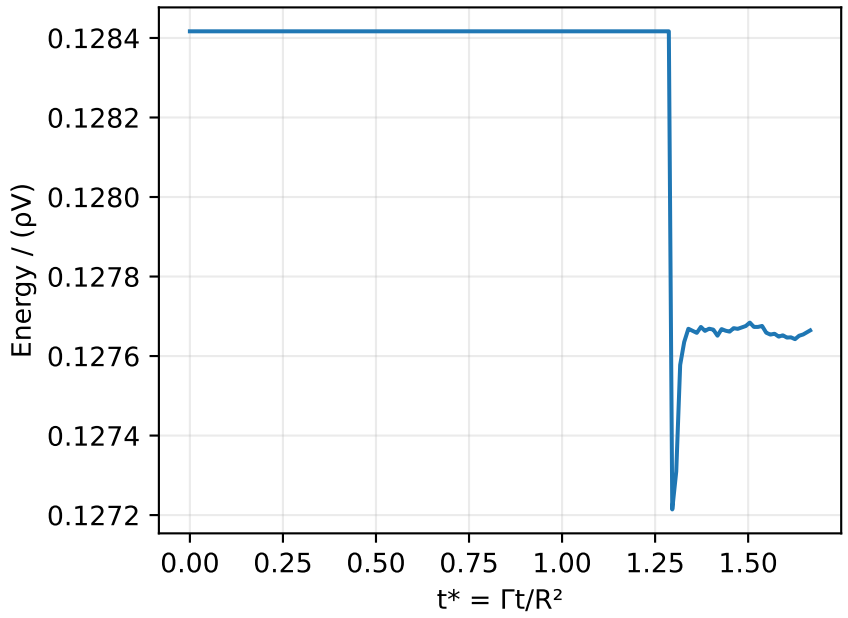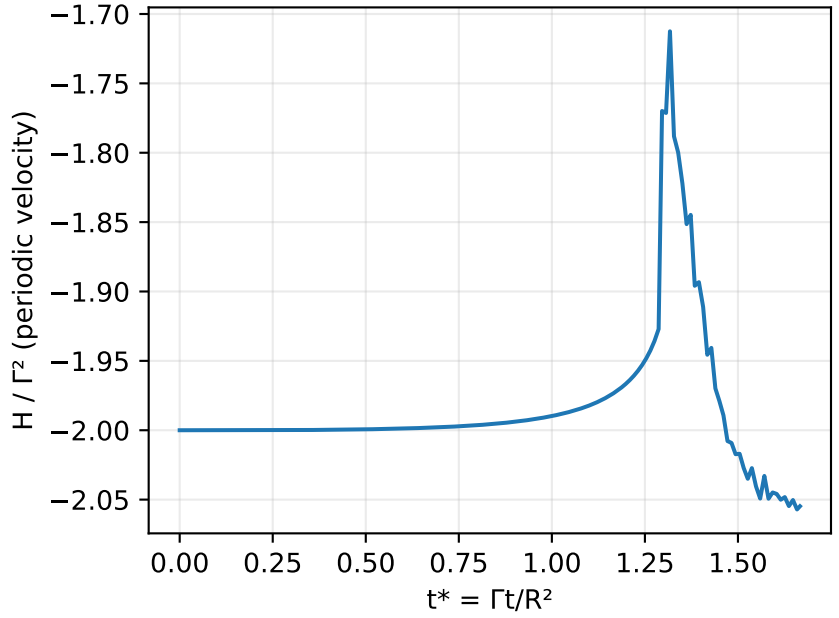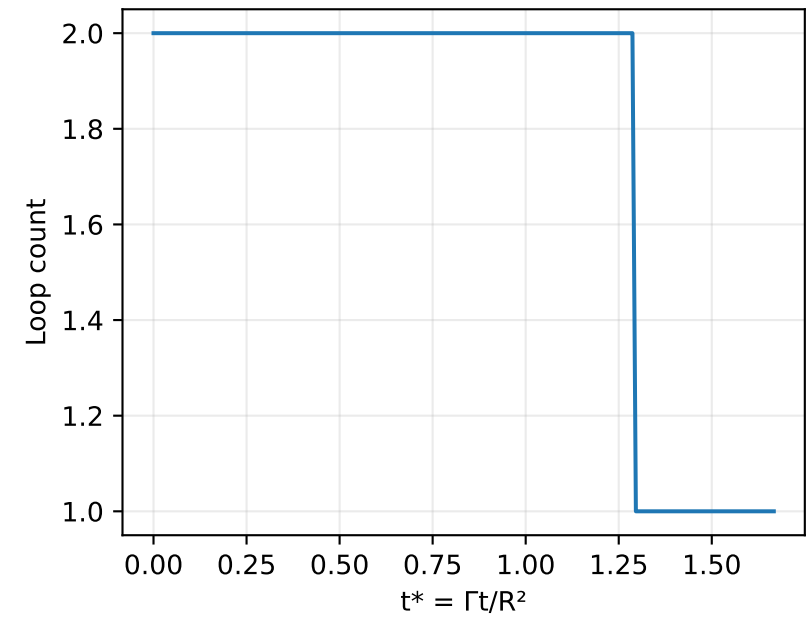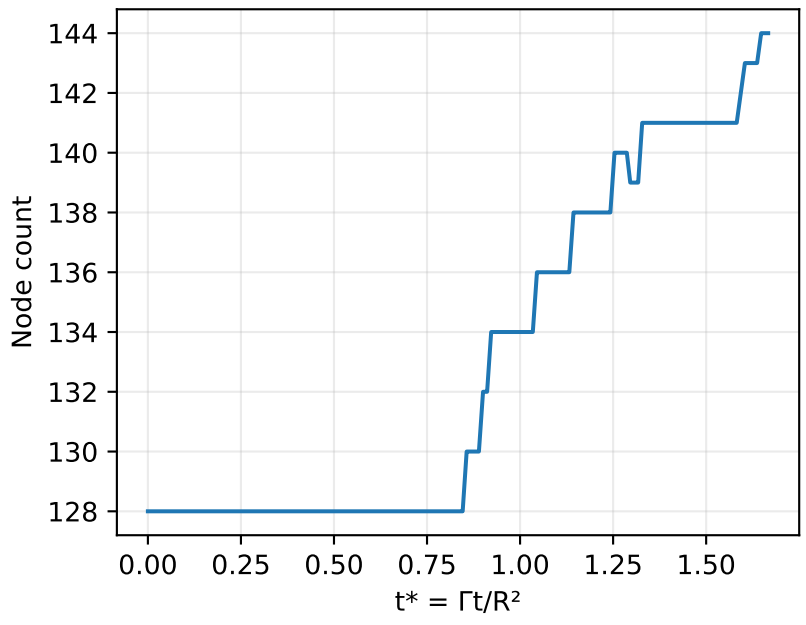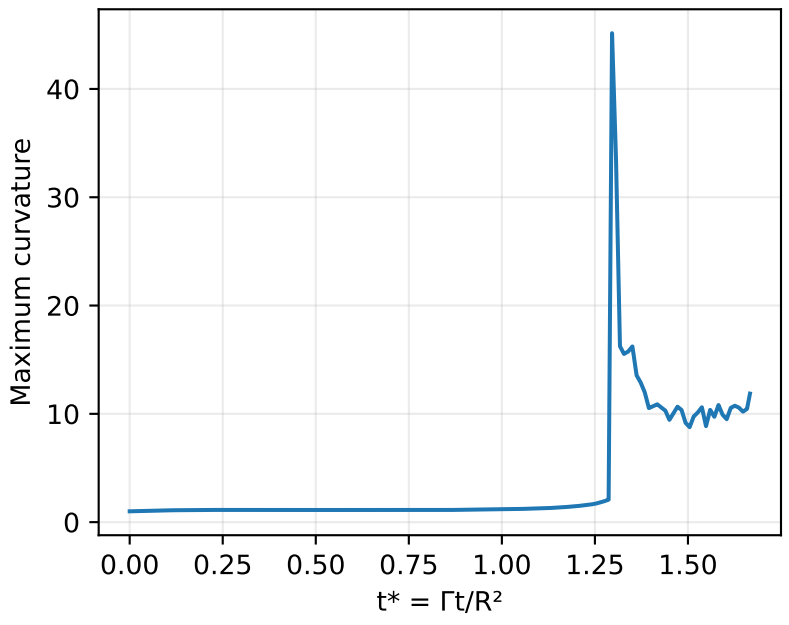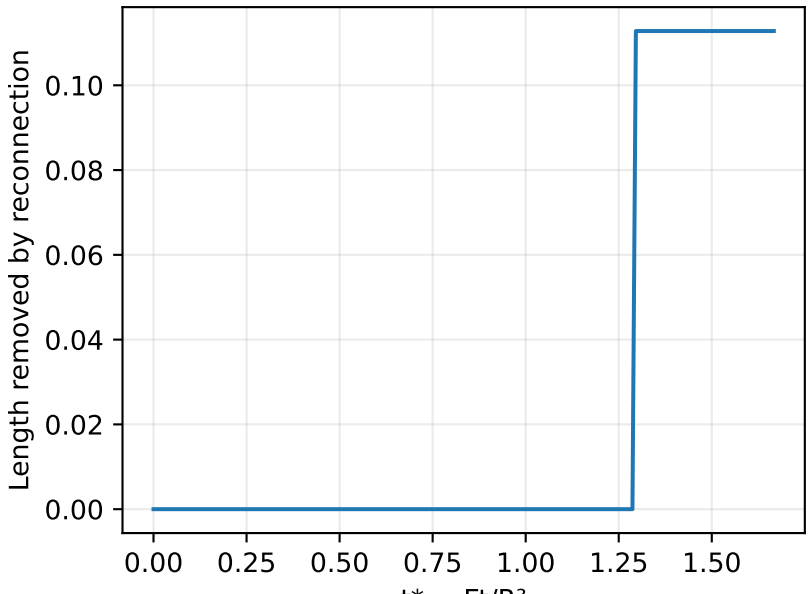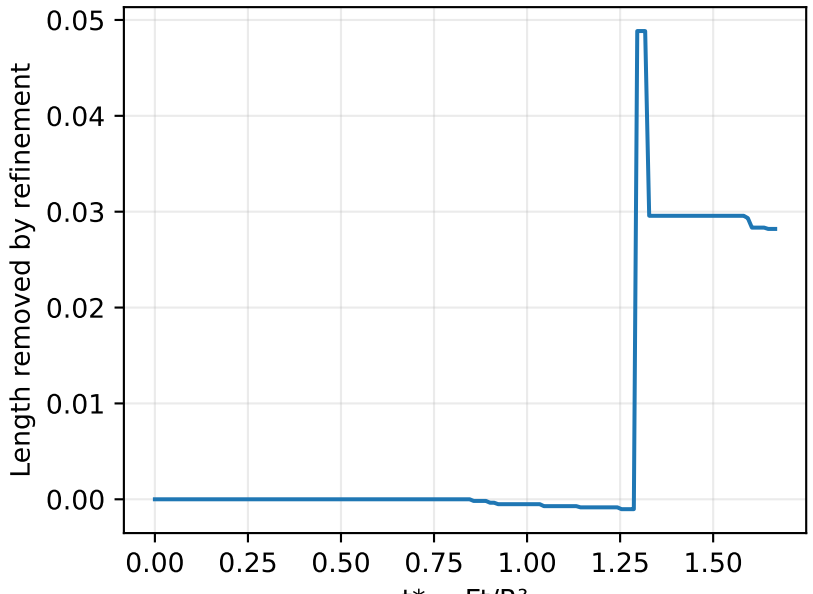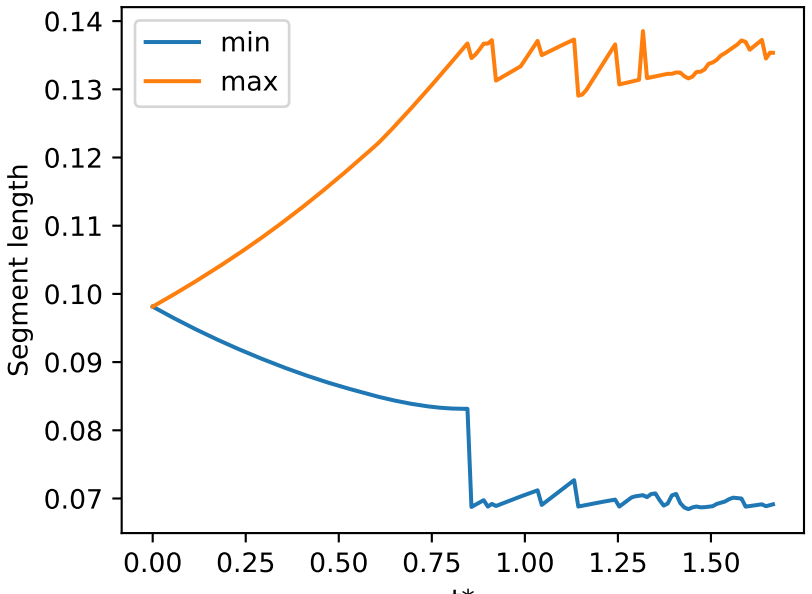hopf_d060 · a=0.001 · no mutual friction